Multiple Programming Data Processing*

B. L. Ryle, *Marc Shiowitz and Associates, Hawthorne, California*

Introduction

In order to put the term "multi-programming" in its proper light, it should be noted that the "program" being executed by a computer system, insofar as the system itself is concerned, consists entirely of the instruction or instructions being executed at any given moment. Considered thus, it is obvious that the system could not care less whether the "current" instruction is somehow associated with the preceding or succeeding instructions. In this sense, multi-programming was being done on the 701 when the program "selected" a tape unit considerably in advance of when the tape was actually to be used, so as to "time-share" the start time of the tape with other operations.

However, when the term "multi-programming" is used today, the implication is that the system referred to has features which simplify the task of the programmer in attempting to maximize the amount of time-sharing possible at any given time. In general, it is not possible to uniquely determine the exact computational sequence of a particular program, since this is usually dependent to some extent on the data being operated on. Thus intra-program time-sharing must be planned by the programmer and can become extremely difficult. An exception exists where multiple runs are to be made on the same program. However, each run may then be treated as a separate program.

Value of Multi-Programming

The value of multi-programming can be summed up very briefly: it is a way of getting more data processed in a given amount of time by given amount of hardware through time-sharing. The amount of data so processed may be taken as a measure of the efficiency of the system; therefore multi-programming is aimed at increasing system efficiency. That time-sharing can "buy" a lot of computing may be seen by looking at the utilization figures for a "typical" system during the execution of a single program with little or no attempt at time-sharing:

Core Memory:	10% of maximum
Average Memory Cell:	0.001% of maximum
Arithmetic Unit:	20% of maximum
Average Control Register:	25–50% of maximum

Cost of Multi-Programming

Multi-programming superimposes an entirely new level of logical complexity on the logical designer and/or programmer. This burden can be shared in many different proportions between the hardware and the program. The factors involved include

(a) additional control logic to facilitate switching between programs,

(b) a supervisory program to monitor the execution of the program set present in the computer at any given time,

(c) new programming systems to provide for the incorporation of the additional program control data required by the control logic of the system and/or the supervisory program.

Design Requirements of a Multi-Program System

The system, to be effective, must be based on the concurrent and mutually coordinated design of the hardware, the supervisory program, and the programming system to be used. These three factors ae so closely related that it is difficult to assert where the responsibility boundaries exist between them. Together, they should provide a system with the following characteristics:

(a) Programs, as entered and/or stored in the system for execution, must be independent of any absolute machine addresses—memory registers, control registers, or other system element addresses.

(b) A system of program priorities and precedence must be included which permits the choice of the appropriate program, with a minimum delay, whenever alternate choices are possible.

(c) A "bookkeeping" system must be provided which keeps track of the current status of each program being executed.

(d) Any register (control or memory), or any other system element not currently being used by a given instruction, must be available for use, if required, by an instruction from another program concurrently being executed.

(e) A system of interrogatory and imperative interrupts must be provided.

(f) A direct communication link must exist between any two system elements which may need to transfer information between themselves. It should never be necessary to use some third element as an intermediate temporary repository for information originating in one element and destined for another.

(g) Sufficient micro-programmability must be incorporated to permit a minimum of redundancy and a maximum of information per instruction, but primarily to permit the utilization of the various registers and logical "blocks" in order to allow one instruction to utilize the idle portion of the duty cycle of another instruction.

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(h) The system must be so organized that little, if any, additional execution time is imposed by the supervision and control necessary to run more than one program.

(i) The amount of data reordering and intra-system data transfer should be minimized.

(j) The job of the programmer must not be made more complex or difficult.

Origin of the Design Requirements

The design requirements noted above originated from the following observations typical of most reasonably large computer installations:

a. The relative proportions of input, arithmetic computation, decision making, data ordering, and output varies over very wide ranges from one program to another, as does both the amount of storage and the execution time.

b. Program check-out and debugging constitutes the single largest application-class of the computer system.

c. The major expense in an installation is the programming, particularly when the machine-time expended in check-out is included.

d. The "priority" workload fluctuates widely.

e. The total workload, including non-priority programs, tends to the capacity of the system (a possible computer corollary of Parkinson's Law).

Types of Multi-Program Systems

A. Central Memory, Multiple Control

1. *Example:* Gamma 60

2. *General Description:* The large central memory is connected to two sets of busses, across which are connected the various other units of the system, such as arithmetic units, logic units, tape units, etc. Each unit, as necessary, sends a request to the central memory for a new instruction. The memory queues the requests if necessary and, from data contained in the request, accesses and transmits the appropriate instruction. Requests for data are treated in a similar fashion. The limiting factor in the system is the rate at which the memory can process the requests for instructions and data.

3. *Comparison Against Design Requirements*

(a) Independence of absolute addressing must be incorporated into the programming system insofar as memory addresses are concerned. Indexing could become very time consuming.

(b) Program priorities and precedence must be incorporated into the supervisory program.

(c) The bookkeeping in this system is done by the operational units.

(d) Any idle operational unit, on the memory, is available for use at any time. The control registers within the units are not separately available.

(e) No interrupt capabilities are provided.

(f) Communication links do not exist between operational units except through the memory.

(g) A form of micro-programming is available, in that each operational unit has its own interpretation of a given instruction code.

(h) The supervisory execution time can well become a major factor in the overall execution time of a program, if it becomes necessary to switch between operating units frequently. Such a switch ties up the memory for a short period, which could cause "lines" to develop.

(i) Intra-system data transfer occurs only between the memory and the registers of one of the operating units.

(j) The programming of the system is rather complex, and it would appear difficult to achieve the degree of time-sharing in practice which might be expected from the theoretical aspects of this system.

4. *Advantages*

a. The number of programs which can be executed simultaneously is limited only by the size of the memory and the number of operating units.

b. If the memory size is modular, such a system can span a very wide range of application requirements.

c. No practical restriction exists on the amount of computation which can be time-shared, except through the limitation of the memory access rate.

5. *Disadvantages*

a. Some queuing is inevitable, and will result in a sub-optimal efficiency level.

b. The reliability of the system is no better than that of the memory, without which the system will not function.

c. Although, in theory, a great deal of computing can be achieved per unit time, the total execution time of a given program can readily become prohibitive.

d. The multi-programming capability is provided by adding complete operating units with independent control capabilities; by thus increasing the number of control registers, the utilization factor for the memory has been increased at the expense of that for the control registers.

B. Multiple Memory, Central Control

1. *Example:* MH 800

2. *General Description:* The major portion of the control logic is concentrated in the central processor. The memory is modular, with each module being capable of independently communicating directly with a single peripheral unit. Each such independent action is initiated by the central processor. All instructions must first pass through the central processor. Each memory module possesses a set of control and index registers which may be associated with a particular one of several programs.

3. *Comparison Against the Design Requirements:*

(a) A rather elaborate system of indexing and relative addressing permits the program to be independent of almost all absolute addressing.

(b) Program priority and precedence must be incorporated into the programming system.

(c) The current status of the various programs being executed is maintained by the independent set of control registers associated with each memory module.

(d) No system element may be utilized until commanded by the central processor. A particular memory module is

always executing a data transfer command, operating with the central processor, or else standing idle.

(e) Direct communication links do not exist between the memory modules. All such communication is via the central processor.

(f) A form of micro-programming exists in the form of diminished addressing, with the diminished portion being supplied by the index register or, more normally, being implied by the location of the command.

(g) Multi-program control is essentially automatic and requires only a small amount of programming.

(h) The programmability index of this system is rather high, with little effort being required to admit the execution of multiple programs.

4. *Advantages*

a. The logic, as mechanized, greatly simplifies the programmer's task.

b. The major system bottleneck—input/output—can be completely time-shared.

5. *Disadvantages*

a. Except when input/output operations are being performed, no two instructions are actually being processed simultaneously.

b. A compute-limited program can have all its input/output requirements absorbed in the compute-time; however, the computation time cannot be further reduced, and conversely for input/output limited programs.

C. Multiple Memory, Multiple Control

1. *Example:* RW 400

2. *General Description:* The central element is a large m x n switching matrix connecting m devices capable of exercising control, to n devices capable of being controlled. The system is highly modular. The two types of "controller" modules provide the fast access memory for the system. Both types are programmable, the difference between them being the presence and lack, respectively, of an arithmetic capability and certain program sequencing capabilities. The controlled modules consist of an arbitrary mix of tape units, drums, etc. Each unit of the system can be connected, through the central exchange, to one, and only one, other module of the system.

3. *Comparison Against Design Requirements*

(a) The system does not possess a basic independence of absolute system addresses, although this can be *partially* achieved through an appropriate programming system.

(b) Program priority and precedence must be incorporated into the supervisory program.

(c) The current status of each program is maintained by the computer or computers executing the program.

(d) Any modules not currently in use are available via the supervisory program. No sub-modular elements may be shared.

(e) A rather elaborate interrupt system is provided.

(f) Direct communication links do not exist between computer modules, nor between processor modules.

(g) No micro-programmability is provided.

(h) Supervision and control can be completely time-shared, provided sufficient system modules are assigned to the task.

(i) The amount of data transfer required in the system can become quite large due to the small computer memory and the inability of the computer to execute a program stored in a processor module without first transferring it.

(j) The programmer's task is greatly complicated by a number of programming inconveniences—lack of index registers, small computer memory, etc.

4. *Advantages*

a. Great reliability, since the system can continue to function in a limited manner so long as any computer module is operable.

b. An extremely wide-range application area.

5. *Disadvantages*

a. An increasingly greater amount of program-shuffling as the size of a given program increases.

b. The absence of index registers or of relative addressing, which tends to increase the length of a given program in an already small memory.

c. A short word with no provision for double-precision operations.

d. The lack of inter-computer and inter-processor communication links, which complicates the problem of generating a supervisory program.

Summary

A multi-program capability in a computer system is a more easily managed approach to improved efficiency through time-sharing. In order to maximize the efficiency it is necessary that the achievable utilization factor for all elements of the system be made to approach unity. This imposes certain design requirements on the combined hardware—programming system—supervisory program complex which have not been fully met by any system announced to date. However, several systems are currently available which partially meet these requirements. Three of these have been described and compared against the criteria. Some of the advantages and disadvantages of each have been mentioned. It should be noted that no system to date has been designed to the specific objectives outlined in this paper. In the Gamma 60, no attempt was made to "share" the internal operational capabilities of any unit except the memory. In the RW-400, the system rather than the individual units have a multiple-program capability, and this capability was a by-product of a rather stringent reliability requirement in the design of the system, rather than being an objective in itself. The MH-800 was designed to provide time-shared input, output and tape operations, with no attempt being made at true multi-programming capabilities.

As digital systems approach the limit in the speed of the individual circuits, the only way in which significant increases in amount of computation per-unit-time can be achieved is through reduction of the "idle time" portion of the duty-cycle for each of the circuits. This is the basic argument for digital systems designed "from scratch" for maximum multi-programmability.